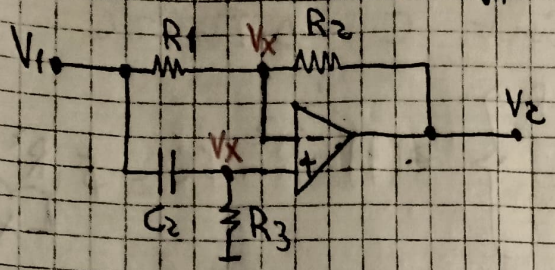
Tarea Semanal #0

① Función Transferencia $\frac{V_2}{V_1}$.




$$\begin{cases} Y_1 = G_1 = \frac{1}{R1} \\ Y_2 = G_2 = \frac{1}{R2} \\ Y_3 = G_3 = \frac{1}{R3} \end{cases}$$

- $V_x(Y_1+Y_2) - V_1 Y_1 - V_2 Y_2 = 0$ ①

- $V_x(Y_3 + \$C_2) - V_1 \$C_2 = 0$

$$V_x = V_1 \frac{\$C_2}{Y_3 + \$C_2} \quad ②$$

② en ①

$$V_1 \left( \frac{\$C_2 (Y_1+Y_2)}{Y_3 + \$C_2} - Y_1 \right) = V_2 Y_2$$

$$V_1 \left( \frac{\cancel{\$Y_1 C_2} + \$Y_2 C_2 - Y_1 Y_3 \cancel{- Y_1 \$C_2}}{Y_3 + \$C_2} \right) = V_2 \; Y_2$$

$$\frac{V_2}{V_1} = \frac{\$C_2 Y_2 - Y_1 Y_3}{\$C_2 Y_2 + Y_2 Y_3} =$$

$$\frac{V_2}{V_1} = \frac{\left(\$ - \frac{Y_1 Y_3}{C_2 Y_2}\right) \cancel{C_2 Y_2}}{\left(\$ + \frac{\cancel{Y_2} Y_3}{C_2 \cancel{Y_2}}\right) \cancel{C_2 Y_2}} = \frac{\$ - \frac{Y_1 Y_3}{C_2 Y_2}}{\$ + \frac{Y_3}{C_2}}$$

$$\frac{V_2}{V_1} = \frac{\$ - \frac{R_2}{C_2 R1 R3}}{\$ + \frac{1}{C_2 R_3}}$$

Si $R_2 = R1$

$$\frac{V_2}{V_1} = \frac{\$ - \overbrace{\frac{1}{C_2 R_3}}^{W_0}}{\$ + \frac{1}{C_2 R_3}}$$

Circuito Pasa Todo

Diagrama de polos y ceros

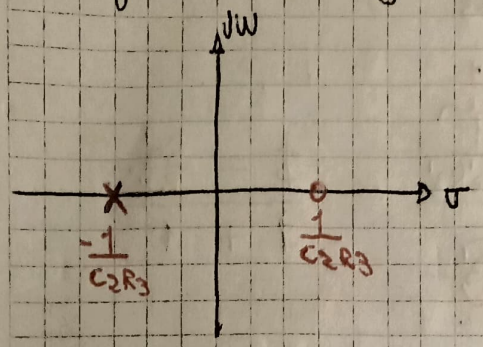
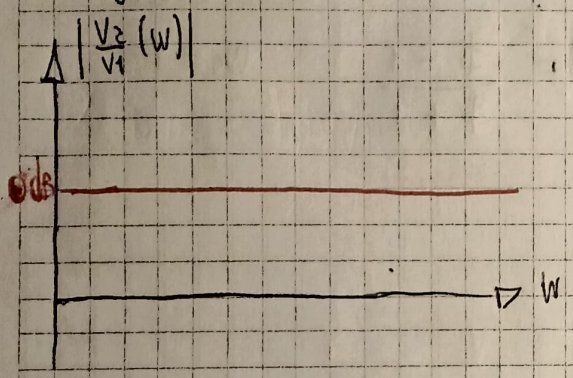
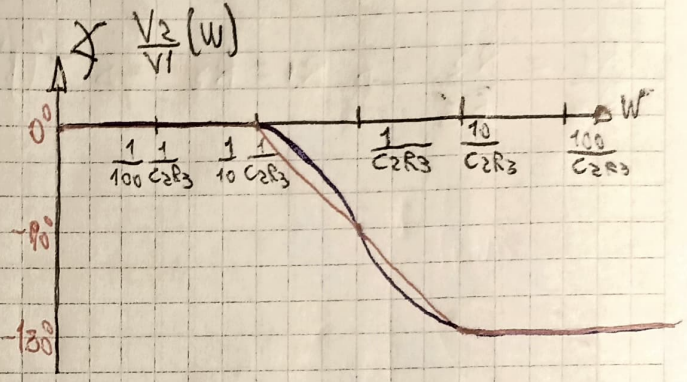



Diagrama de Modulo y fase



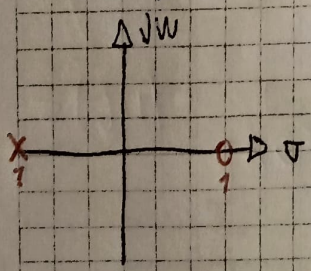




Norma de freq

$w' = w \cdot w_0$

$s' = s \cdot w_0$

$$T(s)\Big|_{s=s'w_0} = \frac{s w_0 - w_0}{s w_0 + w_0} = \frac{s-1}{s+1} \cancel{\frac{w_0}{w_0}}$$




$\Omega_w = w_0 = 1 \wedge w_0 = \frac{1}{C_2R_3} = \Omega_w$

$1 = \frac{1}{C_2R_3} \therefore C_2 = \frac{1}{R_3}$

Normalizacion de impedancia

$\Omega_z = R_3 \therefore R_3' = \frac{R_3}{R_3} = 1 \; ; \; R_2' = \frac{R_2}{R_3} \; ; \; R_1' = \frac{R_1}{R_3} \; ; \; ZC' = \frac{1}{sC \cdot \Omega_z}$

Normalización Conjunta

$$C_2' = C_2 \cdot \Omega_2 \cdot \Omega_w = \cancel{C_2} \cdot \cancel{R_3} \cdot \frac{1}{\cancel{C_2}\cancel{R_3}} = 1$$

$$R_1' = \frac{R_1}{R_3}$$

$$R_2' = \frac{R_2}{R_3}$$

$$R_3' = \frac{R_3}{R_3} = 1$$

$$T'(s) = \frac{s - \frac{R_2'}{R_1'} \cdot \overbrace{\frac{1}{C_2' R_3'}}^{1}}{s + \underbrace{\frac{1}{C_2' R_3'}}_{1}} = \frac{s - \frac{R_2'}{R_1'}}{s + 1}$$

Con $R_2' = R_1'$

$$\frac{R_1}{\cancel{R_3}} = \frac{R_2}{\cancel{R_3}} \rightarrow R_1 = R_2 \rightarrow T'(s) = \frac{s - 1}{s + 1}$$

Para todo $\omega_0 = 1$